

6 Discussion

The empirical picture is sharper than a single reward-hacking narrative. Aggressive PPO produces the clearest localized reward-hacking signal, but the taxonomy also exposes collapse, proxy under-alignment, conservative stagnation, and evaluator-specific disagreement. UP-PPO reduces the density of reward-hacking and evaluator-gaming events in the aggressive condition, but the remaining failures show why mitigation claims should be distributional rather than absolute.

The early-warning results are useful precisely because they are imperfect. The pre-state-only model has discriminative power, suggesting that future failures are not invisible before the transition. Yet average precision remains modest because reward hacking is rare and localized. Operationally, this favors triage: use warning scores to prioritize prompts and checkpoints for expensive human or multi-judge review, not to certify alignment.

The checkpoint-vs-row ablation has the broadest methodological implication. Aggregate metrics remain necessary for reporting, but they can average away exactly the failures that matter. A policy can improve on average while a subset of prompts deteriorates under proxy pressure. RLHF monitoring should therefore include matched prompt-level transitions, not only final or aggregate checkpoint scores.

7 Limitations

This study is deliberately controlled and small scale. The policy family is GPT-2-scale, so the results should be read as evidence about failure-mode observability rather than as a scaling claim about frontier systems. The prompt set is also small: the analysis covers 64 matched prompt identities for row-level transitions, which is enough to demonstrate localization but not enough to estimate population-level failure rates for open-ended deployment.

The analysis is artifact-driven. It consolidates stored rollouts and evaluations rather than running a new multi-seed training campaign. Where possible, we report bootstrap confidence intervals over rows or settings, but these intervals do not replace seed sensitivity. In particular, the mitigation comparison is directionally favorable for UP-PPO, yet the bootstrap intervals for the absolute reductions include zero. A stronger causal claim would require repeated training runs across seeds, larger prompt samples, and fixed judge configurations.

The early-warning models are trained and tested within the same artifact family. The train/test split is stratified by the reward-hacking label but not grouped by prompt id, checkpoint family, or trajectory. The reported ROC-AUC therefore measures within-family discrimination, not transfer to new datasets, model scales, reward models, judge prompts, held-out prompts, or held-out training families. This limitation is especially important because row-level reward hacking is rare and localized; warning models may learn family-specific prompt or trajectory regularities.

The two external judges are useful but not ground truth. The Anthropic/Claude-style and OpenAI-style judges use the same 1–10 helpfulness rubric, but they can still disagree because of model-specific preferences, calibration differences, and sensitivity to response length or wording. We use one judge sample per item and do not include human adjudication or length-controlled judge prompts. Preliminary length diagnostics in Appendix G suggest that verbosity alone is not the main driver, but repeated judge sampling, length-controlled judging, and a small human-rated anchor set would be needed to make that claim robust.

The taxonomy is directional and intentionally simple. It is designed for auditability, but richer temporal models could capture gradual drift, threshold effects, delayed failures, and multi-step causal mechanisms more faithfully. Appendix E shows that classifications are stable for small tolerances but become more conservative as ϵ or the minimum effect-size threshold grows. Finally, MC-dropout uncertainty is used

as a diagnostic and as part of UP-PPO because it was available in the reward-model artifacts, but this paper does not fully validate its calibration or compare it with ensembles, last-layer analytic uncertainty, UWO/WCO-style uncertainty, or adversarial uncertainty penalties. We expect the localization and aggregation results to be less dependent on the uncertainty estimator than the mitigation result, but that expectation remains an empirical claim for future work.

8 Future Work

The highest-return next experiments are straightforward. First, the training runs should be repeated over at least three seeds to separate trajectory-level effects from run-specific artifacts. Second, the diagnostic suite should be run on a stronger small-model setting to test whether the taxonomy survives a change in base model and representation scale. Third, the prompt set should include frontier-style stress tests, including sycophancy, deception, reward-model exploit, and verbosity-exploitation prompts. Fourth, early-warning models should be validated on grouped prompt and trajectory splits, new prompts, different judges, and different reward models, and compared with alternative monitoring frameworks such as reliability diagrams, sequential drift tests, and hazard models. Fifth, UP-PPO should be ablated across uncertainty estimators, dropout rates, numbers of MC samples, and penalty schedules. A small human-adjudicated subset would also anchor judge validity and clarify when evaluator gaming reflects true ambiguity rather than evaluator drift.

9 Conclusion

Reward hacking is best understood as one member of a broader family of RLHF failure modes. By classifying checkpoint transitions, preserving row-level prompt structure, and separating proxy-judge mismatch from judge-judge disagreement, this paper provides a practical diagnostic framework for RLHF monitoring. In the analyzed artifacts, aggressive PPO produces the strongest localized reward-hacking signal; UP-PPO reduces but does not eliminate that signal; pre-transition features anticipate some future failures; and aggregate checkpoint metrics miss localized reward hacking in several settings. The broader lesson is that RLHF evaluation should track how failures emerge, where they localize, and which signals appear before external quality degrades.

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